ANSWERS

1 **(a)** **(i)** change of shape / size / length / dimension C1
when (deforming) <u>force is removed</u>, returns to original shape / size ....A1 [2]

(ii) $L = ke$ ....B1 [1]

(b) 2 e B1
½ k (*allow e.c.f. from extension*) B1
½ e and 2k B1
$\frac{3}{2}$ e (*allow e.c.f. from extension in part 2*) B1
$\frac{2}{3}$ k (*allow e.c.f. from extension*) ....B1 [5]

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2 **(a)** **(i)** returns to original shape / size / length etc. B1
when load / distorting forces / weight / strain is removed ....B1 [2]

(ii) **1** $R = \rho L / A$ ....B1 [1]
2 $E = WL / Ae$ ....B1 [1]

(b) $E = WR / e\rho$ C1
$= (34 \times 0.44) \ / \ (7.7 \times 10^{-4} \times 9.2 \times 10^{-8})$ C1
$= 2.1 \times 10^{11}$ Pa ....A1 [3]

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3 **(a)** ability to do work B1
as a result of a change of shape of an object/stretched etc B1 [2]

(b) work = average force ×distance moved (in direction of the force) B1
either work = ½ × F × x
or work is area under F/x graph which is ½Fx B1
$F = kx$ B1
so work / energy = ½kx^2 A0 [3]

(c) **(i)** spring constant $= \dfrac{3.8}{2.1}$ M1
$= 1.8$ N cm^{-1} A0 [1]

(ii) **1** $\Delta E_P = mg\Delta h$ or $W\Delta h$ C1
$= 3.8 \times 1.5 \times 10^{-2}$
$= 0.057$ J A1 [2]

2 $\Delta E_S = ½ \times 1.8 \times 10^{-2} \ (0.036^2 - 0.021^2)$ M1
$= 0.077$ J A0 [1]

3 work done = 0.077 – 0.057
= 0.020 J A1 [1]
(*allow e.c.f. if* $\Delta E_S > \Delta E_P$)

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4 (a) (i) Young modulus = stress/strain ………………………………………………… C1

data chosen using point in linear region of graph ………………………… M1

Young modulus = $(2.1 \times 10^{8})/(1.9 \times 10^{-3})$

= 1.1×10^{11} Pa ………………………………………………………………... A1 [3]

(ii) This mark was removed from the assessment, owing to a power-of-ten inconsistency in the printed question paper.

(b) area between lines represents energy/area under curve represents energy .. M1

when rubber is stretched and then released/two areas are different ……..... A1

this energy seen as thermal energy/heating/difference represents energy released as heat ……………………………………………………………………… A1 [3]

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5 (a) (i) F / A B1 [1]

(ii) $\Delta L / L$ B1 [1]

(iii) allow $FL / A\Delta L$ B1 [1]

(iv) allow $\rho L / A$ or $\rho(L + \Delta L) / A$ B1 [1]

(b) (i) $\Delta L = FL / EA$

$= (30 \times 2.6) / (7.0 \times 10^{10} \times 3.8) \times 10^{-7}$ M1

$= 2.93 \times 10^{-3}$ m = 2.93 mm A0 [1]

(ii) $\Delta R = \rho\Delta L / A$ C1

$= (2.6 \times 10^{-8} \times 2.93 \times 10^{-3}) / (3.8 \times 10^{-7})$

$= 2.0 \times 10^{-4}\ \Omega$ A1 [2]

(c) change in resistance is (very) small M1

so method is not appropriate A1 [2]

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6 (a) energy = average force × extension B1

$= \frac{1}{2} \times F \times x$ B1

(Hooke's law) extension proportional to (applied) force B1

hence $F = kx$ B1

so $E = \frac{1}{2} kx^2$ A0 [4]

(b) (i) correct area shaded B1 [1]

(ii) 1.0 cm^2 represents 1.0 mJ or correct units used in calculation C1

$E_S = 6.4 \pm 0.2$ mJ A2 [3]

(for answer > ±0.2 mJ but ≤ ±0.4 mJ, then allow 2/3 marks)

(iii) arrangement of atoms / molecules is changed B1 [1]

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7 (a) (i) *stress* is force / area B1 [1]

(ii) *strain* is extension / <u>original</u> length B1 [1]

(b) (i) $E = [F / A] \div [e / l]$ C1

$e = (25 \times 1.7) / (5.74 \times 10^{-8} \times 1.6 \times 10^{11})$ C1

$e = 4.6 \times 10^{-3}$ m A1 [3]

(ii) A becomes $A/2$ or stress is doubled B1

$e \propto l / A$ or substitution into full formula B1

total extension increase is $4e$ A1 [3]

8	**(a)**	clamped horizontal wire over pulley or vertical wire attached to ceiling with mass attached	B1	
		details: reference mark on wire with fixed scale alongside	B1	[2]
	(b)	measure original length of wire to reference mark with metre ruler / tape	(B1)	
		measure diameter with micrometer / digital calipers	(B1)	
		measure initial and final reading (for extension) with metre ruler or other suitable scale	(B1)	
		measure / record mass or weight used for the extension	(B1)	
		good physics method:		
		measure diameter in several places / remove load and check wire returns to original length / take several readings with different loads	(B1)	
		MAX of 4 points	B4	[4]

9	**(a)**		extension is proportional to force (for small extensions)	B1	[1]
	(b)	**(i)**	point beyond which (the spring) does not return to its original length when the load is removed	B1	[1]
		(ii)	gradient of graph = 80 N m^{-1}	A1	[1]
		(iii)	work done is area under graph / ½ *Fx* / ½ kx^2	C1	
			= 0.5 × 6.4 × 0.08 = 0.256 (allow 0.26) J	A1	[2]
	(c)	**(i)**	extension = 0.08 + 0.04 = 0.12 m	A1	[1]
		(ii)	spring constant = 6.4 / 0.12 = 53.3 N m^{-1}	A1	[1]

10	**(a)**	**(i)**	stress = force / (cross-sectional) area	B1	[1]
		(ii)	strain = extension / original length **or** change in length / original length	B1	[1]
	(b)		point beyond which material does not return to the original length / shape / size when the load / force is removed	B1	[1]
	(c)		UTS is the maximum force / original cross-sectional area	M1	
			wire is able to support / before it breaks	A1	[2]
			allow one: maximum stress the wire is able to support / before it breaks		
	(d)	**(i)**	straight line from (0,0)	M1	
			correct shape in plastic region	A1	[2]
		(ii)	only a straight line from (0,0)	B1	[1]
	(e)	**(i)**	ductile: initially force proportional to extension then a large extension for small change in force	B1	
			brittle: force proportional to extension until it breaks	B1	[2]
		(ii)	**1.** does not return to its original length / permanent extension (as entered plastic region)	B1	
			2. returns to original length / no extension **(as no plastic region / still in elastic region)**	B1	[2]

11	**(a)**		Resultant force (and resultant torque) is zero	B1	
			Weight (down) = force from/due to spring (up)	B1	[2]
	(b)	**(i)**	0.2, 0.6, 1.0 s (*one of these*)	A1	[1]
		(ii)	0, 0.8 s (*one of these*)	A1	[1]
		(iii)	0.2, 0.6, 1.0 s (*one of these*)	A1	[1]
	(c)	**(i)**	Hooke's law: extension is proportional to the force (*not mass*)	B1	
			Linear/straight line graph hence obeys Hooke's law	B1	[2]
		(ii)	Use of the gradient (*not just* $F = kx$)	C1	
			$K = (0.4 \times 9.8) / 15 \times 10^{-2}$	M1	
			$= 26(.1)\ \text{N m}^{-1}$	A0	[2]
		(iii)	*either* energy = area to left of line *or* energy = $\frac{1}{2} ke^2$	C1	
			$= \frac{1}{2} \times [(0.4 \times 9.8) / 15 \times 10^{-2}] \times (15 \times 10^{-2})^2$	C1	
			$= 0.294\ \text{J}$ (*allow 2 s.f.*)	A1	[3]
12	**(a)**		E = stress / strain	B1	[1]
	(b)	**(i)**	1. diameter / cross sectional area / radius		
			2. original length	B1	[1]
		(ii)	measure original length with a <u>metre</u> ruler / tape	B1	
			measure the <u>diameter</u> with micrometer (screw gauge) *allow digital vernier calipers*	B1	[2]
		(iii)	energy = $\frac{1}{2} Fe$ or area under graph or $\frac{1}{2} kx^2$	C1	
			$= \frac{1}{2} \times 0.25 \times 10^{-3} \times 3 = 3.8 \times 10^{-4}\ \text{J}$	A1	[2]
	(c)		straight line through origin below original line	M1	
			line through (0.25, 1.5)	A1	[2]
13	**(a)**		when the load is removed then the wire / body object does not return to its original shape / length	B1	[1]
	(b)	**(i)**	stress = force / area	C1	
			$F = 220 \times 10^6 \times 1.54 \times 10^{-6} = 340\ (338.8)\ \text{N}$	A1	[2]
		(ii)	$E = (F \times l) / (A \times e)$	C1	
			$e = (90 \times 10^6) \times 1.75 / (1.2 \times 10^{11}) = 1.31 \times 10^{-3}\ \text{m}$	A1	[2]
	(c)		the stress is no longer proportional to the extension	B1	[1]

14 (a) extension is proportional to force / load	B1	[1]
(b) $F = mg$	C1	
$x = (mg / k) = 0.41 \times 9.81 / 25 = (4.02 / 25)$	M1	
$x = 0.16\,\text{m}$	A0	[2]
(c) (i) weight and (reaction) force from spring (which is equal to tension in spring)	B1	[1]
(ii) F – weight or $0.06 \times 25 = ma$	C1	
$F = 0.2209 \times 25 = 5.52$ (N) or $0.22 \times 25 = 5.5$		
$a = (5.52 - 0.41 \times 9.81) / 0.41$ or $1.5 / 0.41$ and $(5.5 - 4.02)$	C1	
$a = 3.7\ (3.66)\,\text{m s}^{-2}$ gives $3.6\,\text{m s}^{-2}$	A1	[3]
(d) elastic potential energy / strain energy to kinetic energy and gravitational potential energy	B1	
stretching / extension reduces and velocity increases / height increases	B1	[2]

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15 (a) the wire returns to its original length (*not 'shape'*)	M1	
when the load is removed	A1	[2]
(b) energy: N m / kg $\text{m}^2\,\text{s}^{-2}$ and volume m^3	C1	
energy / volume: $\text{kg}\,\text{m}^2\text{s}^{-2} / \text{m}^3$	M1	
energy / volume: $\text{kg}\,\text{m}^{-1}\,\text{s}^{-2}$	A0	[2]
(c) ε has no units	B1	
E: $\text{kg m s}^{-2}\,\text{m}^{-2}$	M1	
units of RHS: $\text{kg}\,\text{m}^{-1}\,\text{s}^{-2}$ = LHS units / satisfactory conclusion to show C has no units	A1	[3]

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16 (a) (i) stress = force / cross-sectional area	B1	[1]
(ii) strain = extension / original length	B1	[1]
(b) (i) E = stress / strain	C1	
$E = 0.17 \times 10^{12}$	C1	
stress $= 0.17 \times 10^{12} \times 0.095 / 100$	C1	
$= 1.6(2) \times 10^{8}\,\text{Pa}$	A1	[4]
(ii) force = (stress × area) $= 1.615 \times 10^{8} \times 0.18 \times 10^{-6}$	C1	
$= 29(.1)\,\text{N}$	A1	[2]

17 **(a)**	(Young modulus / E =) stress / strain	B1	[1]
(b) **(i)**	stress $= F/A$		
	or $= F/(\pi d^2/4)$		
	or $= F/(\pi d^2)$	M1	
	ratio = 4 (or 4:1)	A1	[2]
(ii)	E is the same for both wires (as same material) [e.g. $E_P = E_Q$]	M1	
	strain = stress / E		
	ratio = 4 (or 4:1) [*must be same as* ***(i)***]	A1	[2]
18 **(a)**	add small mass to cause extension then remove mass to see if spring returns to original length	M1	
	repeat for larger masses and note maximum mass for which, when load is removed, the spring does return to original length	A1	[2]
(b)	Hooke's law requires force proportional to extension	B1	
	graph shows a straight line, hence obeys Hooke's law	M1	[2]
(c)	k = force / extension	C1	
	$= (0.42 \times 9.81) / [(30 - 21.2) \times 10^{-2}]$	C1	
	$= 47\ (46.8)\ \text{N m}^{-1}$	A1	[3]
19 **(a)** **(i)**	solid: (molecules) vibrate	B1	
	no translational motion / fixed position, liquid: translational motion	B1	[2]
(ii)	gas: molecules have random (and translational) motion	B1	[1]
(b) **(i)**	ductile: straight line through origin then curving towards x-axis	B1	[1]
(ii)	brittle: straight line through origin with no or negligible curved region	B1	[1]
(c)	similarity: obey Hooke's law / $F \propto x$ or have elastic regions	B1	
	difference: brittle no or (very) little plastic region		
	ductile has (large(r)) plastic region	B1	[2]
20 **(a)** **(i)**	two sets of co-ordinates taken to determine a constant value (F/x)	M1	
	F/x constant hence obeys Hooke's law	A1	[2]
or	gradient calculated and one point on line used	(M1)	
	to show no intercept hence obeys Hooke's law	(A1)	
(ii)	gradient or one point on line used e.g. $4.5 / 1.8 \times 10^{-2}$	C1	
	(k =) $250\ \text{N m}^{-1}$	A1	[2]
(iii)	work done or E_P = area under graph or $\frac{1}{2}Fx$ or $\frac{1}{2}kx^2$	C1	
	$= 0.5 \times 4.5 \times 1.8 \times 10^{-2}$ or $0.5 \times 250 \times (1.8 \times 10^{-2})^2$	C1	
	$= 0.041\ (0.0405)\ \text{J}$	A1	[3]
(b)	KE $= \frac{1}{2}mv^2$		
	$\frac{1}{2}mv^2 = 0.0405$ or KE = 0.0405 (J)	C1	
	($v = [2 \times 0.0405 / 1.7]^{1/2}$ =) $0.22\ (0.218)\ \text{m s}^{-1}$	A1	[2]

21	**(a)**	extension is proportional to force (for small extensions)	B1	[1]
	(b) (i)	point beyond which (the spring) does not return to its original length when the load is removed	B1	[1]
	(ii)	gradient of graph = 80 N m^{-1}	A1	[1]
	(iii)	work done is area under graph / ½ *Fx* / ½ kx^2	C1	
		= 0.5 × 6.4 × 0.08 = 0.256 J (allow 0.26 J)	A1	[2]

22	**(a)**	force/load is proportional to extension/compression (provided proportionality limit is not exceeded)	B1	[1]
	(b) (i)	$k = F/x$ *or* k = gradient	C1	
		k = 600 N m^{-1}	A1	[2]
	(ii)	(W =) ½kx^2 *or* (W =) ½Fx *or* (W =) area under graph	C1	
		(W =) 0.5 × 600 × (0.040)^2 = 0.48 J *or* (W =) 0.5 × 24 × 0.040 = 0.48 J	A1	[2]
	(iii) 1.	(E_K =) ½mv^2	C1	
		= ½ × 0.025 × 6.0^2		
		= 0.45 J	A1	[2]
	2.	(work done against resistive force =) 0.48 – 0.45 [= 0.03(0) J]	C1	
		average resistive force = 0.030 / 0.040	C1	
		= 0.75 N	A1	[3]
	(iv)	efficiency = [useful energy out / total energy in] (×100)	C1	
		= [0.45 / 0.48] (×100)		
		= 0.94 *or* 94%	A1	[2]

23	(a)	E = stress / strain **or** $(F/A)/(e/l)$	**C1**
		= [gradient × 3.5] / [π × (0.19 × 10^{-3})^2]	**C1**
		e.g. E = [{(40 – 5) / ([11.6 – 3.2] × 10^{-3})} × 3.5] / [π × (0.19 × 10^{-3})^2] **or** [4170 × 3.5] / [π × (0.19 × 10^{-3})^2]	
		E (= 1.3 × 10^{11}) = 0.13 TPa (*allow answers in range 0.120–0.136 TPa*)	**A1**
	(b)	a larger <u>range</u> of F required **or** <u>range</u> greater than 35 N	**B1**

24 (a)	(strain =) extension / original length	B1
(b) (i)	$E = \sigma / \varepsilon$	C1
	maximum stress $= 2.1 \times 10^{11} \times 4.0 \times 10^{-4}$	A1
	$= 8.4 \times 10^{7}$ Pa	
(ii)	$\sigma = F/A$	C1
	minimum area $= 8.0 \times 10^{3} / 8.4 \times 10^{7}$	A1
	$= 9.5 \times 10^{-5}\ \text{m}^2$	

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25 (a)	$p = 1000 \times 9.81 \times 7.0 \times 10^{-2}$ **or** $1000 \times 9.81 \times 1.9 \times 10^{-2}$	C1
	$\Delta p = 1000 \times 9.81 \times (7.0 \times 10^{-2} - 1.9 \times 10^{-2})$ **or** $686 - 186$	A1
	$= 500$ Pa	
(b)	$F = pA$ **or** $(\Delta)F = \Delta p \times A$	C1
	upthrust $= 500 \times (5.1 \times 10^{-2})^2 = 1.3$ N **or** upthrust $= (686 - 186) \times (5.1 \times 10^{-2})^2 = 1.3$ N **or** upthrust $= 1000 \times 9.81 \times 5.1 \times 10^{-2} \times (5.1 \times 10^{-2})^2 = 1.3$ N	A1
(c)	force $= 4.0 - 1.3$	A1
	$= 2.7$ N	
(d)	extension/x/e $= 2.7 / 30$	C1
	$= 0.09$ (m) **or** 9 (cm)	C1
	height above surface $= 9 - 7$	A1
	$= 2$ cm	
(e) (i)	mass $= 4.0 / 9.81$	C1
	acceleration $= 2.7 / (4.0 / 9.81)$	A1
	$= 6.6\ \text{m s}^{-2}$	
(ii)	viscous force increases (and then becomes constant)	M1
	(weight and upthrust constant so) acceleration decreases (to zero)	A1

26 (a)	$\rho = m/V$	**C1**
	$= (560/9.81)/(1.2 \times 0.018)$	**A1**
	$= 2600\,\text{kg m}^{-3}$	
(b)	$(\triangle)p = 940 \times 9.81 \times 1.2$	**C1**
	(upthrust =) $940 \times 9.81 \times 1.2 \times 0.018 = 200\,\text{N}$	**A1**
(c) (i)	tension $= 560 - 200$	**A1**
	$= 360\,\text{N}$	
(ii)	$P = Fv$	**C1**
	$= 360 \times 0.020$	**A1**
	$= 7.2\,\text{W}$	
(d) (i)	upthrust decreases	**B1**
	tension (in wire) increases	**M1**
	power (output of motor) increases	**A1**
(ii)	there is work done (on the cylinder) by the upthrust	**B1**
or	GPE of oil decreases (as it fills the space left by cylinder and so total energy is conserved)	

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27 (a)	(Young modulus =) stress / strain	**B1**
(b) (i)	$k = F/\triangle L$ **or** 1 / gradient	**C1**
	$= 90 \times 10^{3}/(2 \times 10^{-3})$ (*or other point on line*)	**A1**
	$= 4.5 \times 10^{7}\,\text{N m}^{-1}$	
(ii)	$E = ½F\triangle L$ **or** $E = ½k(\triangle L)^2$	**C1**
	$= ½ \times 90 \times 10^{3} \times 2 \times 10^{-3}$ **or** $½ \times 4.5 \times 10^{7} \times (2 \times 10^{-3})^2$	**C1**
	$= 90\,\text{J}$	**A1**
(c)	straight line starting from (0, 150) and below original line	**M1**
	line ends at (90, 147)	**A1**

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28 (a)	current temperature (**allow** amount of substance, luminous intensity) *any two correct answers, 1 mark each*	**B2**
(b) (i)	$W = 2 \times (150 \times \sin 17°)$ **or** $2 \times (150 \times \cos 73°)$	**C1**
	$W = 88$ N	**A1**
(ii)	**1.** $\sigma = F/A$	**C1**
	$= 150/(7.5 \times 10^{-6})$	**A1**
	$= 2.0 \times 10^{7}$ Pa	
	2. $\varepsilon = \sigma / E$	**C1**
	$= 2.0 \times 10^{7}/(2.1 \times 10^{11})$	**A1**
	$= 9.5 \times 10^{-5}$	

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29 (a)(i)	$p = mv$	**C1**
	$= 0.2(00) \times 6.(00) \times \sin 60(.0)°$ **or** $0.2(00) \times 6.(00) \times \cos 30(.0)°$	**A1**
	$= 1.04$ kg m s^{-1}	
(ii)	$0.300 \times v_x \times \sin 60.0° = 1.04$	**A1**
	$v_x = 4.00$ m s^{-1}	
(iii)	$0.30 \times 4.0 \times \cos 60°$ **or** $0.20 \times 6.0 \times \cos 60°$ **or** $(0.30 + 0.20)v$ **or** $0.50v$	**C1**
	$0.30 \times 4.0 \times \cos 60° + 0.20 \times 6.0 \times \cos 60° = (0.30 + 0.20)v$ **or** $0.50v$	**A1**
	so $v = 2.4$ m s^{-1}	
(b) (i)	$E = ½mv^2$	**C1**
	$½ \times 0.50 \times 2.4^2 = ½ \times 72 \times x^2$	**C1**
	$x = 0.20$ m	**A1**
(ii)	**1.** straight line from the origin sloping upwards	**B1**
	2. line drawn from a positive value of E_k at $x = 0$ to a positive value of x at $E_k = 0$	**M1**
	line has an increasing downwards slope	**A1**

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30 (a) Hooke's (law) **B1**

(b)(i) $\sigma = F / A$ **C1**

$= 36 / (4.1 \times 10^{-7})$ **A1**

$= 8.8 \times 10^{7}$ Pa

(ii) Young modulus = σ / ε **or** $F / A\varepsilon$ **C1**

$\varepsilon = 8.8 \times 10^{7} / (1.7 \times 10^{11})$ **A1**

$= 5.2 \times 10^{-4}$

(c) $R = \rho L / A$ **C1**

$\Delta R = \rho \Delta x / A$ **C1**

$= 3.7 \times 10^{-7} \times 0.12 \times 10^{-3} / (4.1 \times 10^{-7})$

$= 1.1 \times 10^{-4}\ \Omega$ **A1**

(d) remove the force/F and wire returns to original length **B1**

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31 (a) for a body in (rotational) equilibrium **B1**

sum/total of clockwise moments about a point = sum/total of anticlockwise moments about the (same) point **B1**

(b)(i) $(W \times 0.45)$ **or** (19×1.3) **or** $(W \times 1.85)$ **or** (22×2.6) **C1**

$(W \times 0.45) + (19 \times 1.3) + (W \times 1.85) = (22 \times 2.6)$ so $W = 14$ N **A1**

(ii) magnitude $= 19 + 14 + 14 - 22$ **A1**

$= 25$ N

direction: vertically upwards **A1**

(c)(i) the extension is zero when the force is zero **B1**

graph is a straight line and (so) Hooke's law obeyed **B1**

(ii) $k = F / x$ **or** k = gradient **C1**

e.g. $k = 60 / (1.00 - 0.25)$ **A1**

$k = 80$ N m^{-1}

(iii) area shaded below graph line between $L = 0.25$ m and $L = 0.75$ m **B1**

32.

(a)(i)	(stress =) force / cross-sectional area
(a)(ii)	(strain =) extension / original length
(b)(i)	$E = FL / Ax$
	$= GL / A$
(b)(ii)	straight line from origin above the original line
	line ends at point (4 small squares, F_1).
(b)(iii)	**1.** shaded area below the graph line and between the two vertical dashed lines
	2. remove the force/F/F_2 and the wire goes back to original length/zero extension
(b)(iv)	values have a large range

33.

(a)(i)	$F = kx$
	$F_1 = 800 \times 0.045$ $= 36$ N
(a)(ii)	(E =) $\frac{1}{2}kx^2$ **or** $\frac{1}{2}Fx$ **or** area under graph
	$\frac{1}{2} \times 800 \times (0.045)^2$ **or** $\frac{1}{2} \times 36 \times 0.045 = 0.81$ (J)
(b)(i)	efficiency = $(0.72 / 0.81) \times 100$ $= 89\%$
(b)(ii)	$E = \frac{1}{2}mv^2$
	$p = mv$
	$0.72 = \frac{1}{2} \times 0.020 \times v^2$ and $p = 0.020 \times v$ $p = 0.17$ N s
(c)(i)	$(\Delta)E = mg(\Delta)h$
	$h = 0.60 / (0.020 \times 9.81) = 3.1$ m
(c)(ii)	$F = (0.72 - 0.60) / 3.1$
	$= 0.039$ N
(c)(iii)	resultant force on ball is less (than that with air resistance) so time (taken) is more (than T)

34.

(a)	compression/extension is proportional to force (provided limit of proportionality is not exceeded)
(b)	(E) = ½Fx **or** ½kx^2 **or** area under graph
	= ½ × 8 × 16 × 10^{-2} = 0.64 (J) **or** = ½ × 50 × (16 × 10^{-2})^2 = 0.64 (J)
(c)(i)	(E) = ½mv^2
	0.64 = ½ × 0.076 × v^2 v = 4.1 m s^{-1}
(c)(ii)	(Δ)(E) = $mg(\Delta)h$
	= 0.076 × 9.81 × 0.24 (= 0.18 (J))
	kinetic energy = 0.64 – 0.18 = 0.46 J
c)(iii)	v = 4.1 m s^{-1}
(d)	$W = Fs$
	d = 0.30 + (2π × 0.12) + 0.25 (= 1.3 m)
	F = 0.23 / 1.3 = 0.18 N

35.

(a)	Hooke's (law)
(b)(i)	$k = F / x$ or k = gradient e.g. k = 7.0 / 5.0 × 10^{-2}
	= 140 N m^{-1}
(b)(ii)	E= ½ F x or E = ½ k x^2 or E = area under graph = ½ × 5.6 × 4.0 × 10^{-2} or ½ × 140 × (4.0 × 10^{-2})^2
	= 0.11 J
(c)(i)	(upthrust =) 6.20 – 5.60 = 0.60 (N)
(c)(ii)	$\Delta p = \Delta F / A$ = 0.60 / 1.2 × 10^{-3}
	= 500 Pa

(c)(iii)	$(\Delta)p = \rho g(\Delta)h$ $\rho = 500 / (9.81 \times 5.8 \times 10^{-2})$
	$= 880 \text{ kg m}^{-3}$
(d)(i)	(upthrust) increases
(d)(ii)	(extension) decreases

36.

(a)	force $\times$ distance
	perpendicular distance of (line of action of) force from the point
(b)(i)	distance moved by pointer = 123 – 86 (= 37 mm)
	(extension =) $37 \times (1.8 / 52.6) = 1.3$ (mm) **or** sin or tan $\theta = 37 / 526$ (so $\theta = 4.0°$ so extension =) sin or tan $\theta \times 18 = 1.3$ (mm)
(b)(ii)	moment $= 0.472 \times 9.81 \times 6.2 \times 10^{-2}$
	$= 0.29$ N m
(b)(iii)	$(\Delta)F \times 1.8 \times 10^{-2} = 0.29$
	$\Delta F = 16$ N
(b)(iv)	$k = F / x$
	$= 16 / (1.3 \times 10^{-3})$ $= 1.2 \times 10^{4} \text{ N m}^{-1}$

37.

(a)(i)	$\sigma = F / xy$
(a)(ii)	$\varepsilon = (z - w) / w$
(a)(iii)	$E = \sigma / \varepsilon$
	$= Fw / xy(z - w)$
(b)(i)	extension = 2.2 mm (*allow* 2.0–2.4 mm)
(b)(ii)	strain energy = area under graph/line **or** $\frac{1}{2}Fx$ **or** $\frac{1}{2}kx^2$
	$= \frac{1}{2} \times 120 \times 1.4 \times 10^{-3}$ **or** $\frac{1}{2} \times 8.6 \times 10^{4} \times (1.4 \times 10^{-3})^2$
	$= 0.084$ J
(b)(iii)	(some of the) deformation of the wire is plastic/permanent/not elastic **or** wire goes past the elastic limit/enters plastic region
	energy (that cannot be recovered) is dissipated as thermal energy/becomes internal energy

38.

(a)	$k = F / \Delta L$ **or** F / x **or** gradient
	= e.g. 30 / (0.60 – 0.20) = 75 N m^{-1}
(b)	$E = \frac{1}{2}F\Delta L$ **or** $\frac{1}{2}Fx$ **or** $\frac{1}{2}k(\Delta L)^2$ **or** $\frac{1}{2}kx^2$ **or** area under graph
	$= \frac{1}{2} \times 15 \times 0.20$ **or** $\frac{1}{2} \times 75 \times 0.20^2$
	= 1.5 J

39.

(a)(i)	$t = 1.8 / 4.9$ = 0.37 s
(a)(ii)	$v = u + at$ = 9.81 × 0.37
	= 3.6 m s^{-1}
(a)(iii)	$v^2 = 3.6^2 + 4.9^2$
	$v = 6.1$ m s^{-1}
	$\theta = \tan^{-1}(3.6 / 4.9)$ = 36°
(b)(i)	$\rho = m / V$
	$V = \frac{4}{3}\pi r^3$
	$\rho = 0.017 / [\frac{4}{3}\pi \times (0.016 / 2)^3]$ = 7900 kg m^{-3}
(b)(ii)	(E =) ½mv^2
	(E =) ½ × 0.017 × 4.9^2 = 0.20 (J)

(b)(iii)	$k = F / x$ **or** k = gradient
	e.g. $k = 6.4 / 10 \times 10^{-2}$ $= 64$ N m^{-1} (*allow 63–65 N m^{-1}*)
(b)(iv)	$E = ½kx^2$ **or** $E = ½Fx$ **and** $F = kx$
	$x_0 = [(2 \times 0.20) / 64]^{0.5}$ $= 0.079$ m or 0.080 m
(c)(i)	same elastic potential energy / same (initial) kinetic energy and (polystyrene ball has) smaller mass (so greater speed) **or** same (average) force and (polystyrene ball has) smaller mass, (so greater average acceleration so greater speed)
(c)(ii)	(for the polystyrene ball there is) less (average vertical) acceleration / smaller (average vertical component of) resultant force (so takes longer time to reach ground)

40.

(a)	$E = ½mv^2$
	$= ½ \times 0.25 \times 2.3^2$ $= 0.66$ J
(b)	$E = ½kx^2$ **or** $E = ½Fx$ **and** $F = kx$
	$x = [(2 \times 0.66) / 420]^{0.5}$ $= 0.056$ m
	or
	$E = ½kx^2$
	$½mv^2 = ½kx^2$ $x = (0.25 \times 2.3^2 / 420)^{0.5}$ $= 0.056$ m
(c)(i)	(p =) 0.25×2.3 **or** 0.25×1.5
	change in momentum = 0.25 (2.3 + 1.5) = 0.95 N s
(c)(ii)	resultant force = 0.95 / 0.086 or 0.25 × (2.3 + 1.5) / 0.086 = 11 N
(d)	curved line from the origin with an increasing gradient

41.

(a)	sum of CW moments = sum of ACW moments
	about the same point for (an object in rotational) equilibrium
(b)	moment = 0.3(0) $\times$ 0.29 cos 40° **or** 0.3(0) $\times$ 0.222
	= 0.067 N m
(c)(i)	$k = F / x$ **or** k = gradient
	e.g. $k = 21 / 10 \times 10^{-3}$ $k = 2100\ \text{N m}^{-1}$
(c)(ii)	$V_{(\text{sphere})} = \frac{4}{3} \times \pi \times (0.0480)^3$
	$F = \rho g V$ (upthrust =) $1000 \times 9.81 \times (\frac{4}{3} \times \pi \times (0.048)^3) \times 0.26(0) = 1.18$ (N)
(c)(iii)	1.18 $\times$ 0.29 **or** 0.30 $\times$ 0.29 **or** $F \times 0.017$
	$(1.18 \times 0.29) = (0.30 \times 0.29) + (F \times 0.017)$ F = 15 N
(c)(iv)	$E_{(P)} = ½kx^2$ **or** $E_{(P)} = ½Fx$
	$x = F / k = 15 / 2100$ **or** x determined from graph for F = 15.0 N $E_P = ½ \times 2100 \times (15 / 2100)^2$ **or** $E_P = ½ \times 15 \times (15 / 2100)$ E_P = 0.054 J
(d)	the sphere has gained gravitational potential energy

42.

(a)	Hooke's (law)
(b)	$k = F / x$ **or** k = gradient
	= e.g. 12.0 / (0.240 – 0.08) = 75 N m^{-1}
(c)	$E = \frac{1}{2}Fx$ **or** $E = \frac{1}{2}kx^2$ **or** E = area under graph
	$E = \frac{1}{2} \times 6.0 \times 0.080$ **or** $\frac{1}{2} \times 75 \times 0.08^2$ = 0.24 J

43.

(a)	$E_{(P)} = \frac{1}{2}kx^2$ **or** $E_{(P)} = \frac{1}{2}Fx$ **and** $F = kx$
	$0.048 = \frac{1}{2} \times k \times (2.1 \times 10^{-2})^2$ $k = 220\ \text{N m}^{-1}$
(b)	$E_{(K)} = \frac{1}{2}mv^2$
	$0.048 = \frac{1}{2} \times 7.5 \times 10^{-3} \times v^2$ $v = 3.6\ \text{m s}^{-1}$
(c)(i)	$(\Delta)E = mg(\Delta)h$
	$0.039 = 7.5 \times 10^{-3} \times 9.81 \times (\Delta)h$ $\Delta h = 0.53\ \text{m}$
(c)(ii)	$F \times 0.53 = 0.048 - 0.039$
	$F = 0.02\ \text{N}$
(d)	sketch: curved line from the origin
	curved line has increasing gradient

44.

(a)(i)	spring constant
(a)(ii)	area represents the work done (to extend the wire)
	work done (to extend the wire) is equal to elastic potential energy
(b)(i)	$x_G = 0.39$ mm **and** $x_H = 0.29$ mm
(b)(ii)	$E = \frac{1}{2}Fx$ **or** $E = \frac{1}{2}kx^2$ **and** $F = kx$ **or** E = area under graph
	for G: $E = \frac{1}{2} \times 2.0 \times (0.39 \times 10^{-3})$ **or** $\frac{1}{2} \times 5.1 \times 10^3 \times (0.39 \times 10^{-3})^2$ for H: $E = \frac{1}{2} \times 2.0 \times (0.29 \times 10^{-3})$ **or** $\frac{1}{2} \times 6.9 \times 10^3 \times (0.29 \times 10^{-3})^2$ $E_P = 3.9 \times 10^{-4} + 2.9 \times 10^{-4}$ $= 6.8 \times 10^{-4}$ J
(b)(iii)	$E = FL / Ax$ **or** stress / strain = FL / Ax
	$A_G / A_H = 1 \times (0.29 \times 10^{-3}) / [1.5 \times (0.39 \times 10^{-3})]$
	ratio = 0.50